

Optimal Wind Power Uncertainty Intervals for Electricity Market Operation

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Abstract—It is important to select an appropriate uncertainty level of the wind power forecast for power system scheduling and electricity market operation. Traditional methods hedge against a predefined level of wind power uncertainty, such as a specific confidence interval or uncertainty set, which leaves the questions of how to best select the appropriate uncertainty levels. To bridge this gap, this paper proposes a model to optimize the forecast uncertainty intervals of wind power for power system scheduling problems, with the aim of achieving the best trade-off between economics and reliability. Then, we reformulate and linearize the models into a mixed integer linear programming (MILP) without strong assumptions on the shape of the probability distribution. In order to invest the impacts on cost, reliability, and prices in a electricity market, we apply the proposed model on a two-settlement electricity market based on a six-bus test system and on a power system representing the U.S. state of Illinois. The results show that the proposed method can not only help to balance the economics and reliability of the power system scheduling, but also help to stabilize the energy prices in electricity market operation.

Index Terms—Electricity market, MILP, optimal uncertainty interval, power system scheduling, wind power uncertainty.

NOMENCLATURE

Indices:

i, j, l, t, b, m	Indices of thermal units/ wind farms/ transmission lines/ hours/ buses/ linearization segments
s	Indices scenarios for $\{0, LB, UB\}$, where 0 is the central scenario, UB is the upper bound scenario, LB is the lower bound scenario.

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Variables:

I_{it}	Commitment of unit i in hour t .
α_{it}, β_{it}	Start up/ shut down of unit i in hour t .
$P_{G, it}^s, P_{W, jt}^s$	Power dispatch of thermal units i / wind farms j in scenario s , hour t [MW].
$\omega_{jt}^{LB}, \omega_{jt}^{UB}$	Lower and upper bound of wind power forecast uncertainty interval of wind farm j in hour t [p.u.].
SU_{it}, SD_{it}	Startup/shutdown cost, unit i in hour t [\$].
F_{ci}	Cost curve of thermal units i .
$P_{D, bt}^{ens}$	Energy not served at bus b in hour t [MW].
$P_{W, jt}^{wc}$	Wind curtailment at wind farm j in hour t [MW].
P_{jt}^{eeens}	Expected energy not served caused by wind farm j in hour t [MW].
P_{jt}^{ewc}	Expected wind curtailment at wind farm j in hour t [MW].
ξ_{mjt}, ζ_{mjt}	Dummy variables for linearization of wind farms j for segment m in hour t .
$R_{G, it}^{up}, R_{G, it}^{dn}$	Up and down reserve provided by unit i in hour t [MW].

Constants:

$\pi^c, \pi^{st}, \pi^{sd}$	Prices of fuel/thermal units start up/ thermal units shut down [\$/MWh].
π^{wc}, π^{ewc}	Prices of wind power curtailment/ expected wind power curtailment [\$/MWh].
π^{ens}, π^{eeens}	Prices of energy not served/ expected energy not served [\$/MWh].
$P_{D, bt}^f$	System demand at bus b in hour t [MW].
$P_{W, jt}^f$	Wind power forecast of wind farm j in hour t [MW].
$P_{Gi, \min}, P_{Gi, \max}$	Minimum/maximum capacity of unit i [MW].
P_{Wj}^{inst}	Installed capacity power of wind farm j [MW].
T_i^{on}, T_i^{off}	Minimum ON/OFF time of unit i [h].
RU_i, RD_i	Ramp up/down limits of unit i [MW/h].
RSU_i, RSD_i	Ramp startup/shutdown limits of unit i [MW/h].
$PL_{l, \max}$	Power flow limit of line l in hour t [MW].
$\hat{R}_{req, t}^{up}, \hat{R}_{req, t}^{dn}$	Up and down reserve requirement in hour t [MW].

NT, NG, NW, NB Number of hours/ thermal units/ wind farms/ buses.
 G Shift factor matrix.

I. INTRODUCTION

THE increasing penetration of wind power brings significant challenges to the operation of electric power systems. To better represent the wind power uncertainty, various representations of wind power forecast uncertainty have been proposed, such as probability distributions [1]–[2], scenarios [3], intervals [4]–[6] and uncertainty sets [7]. Correspondingly, substantial efforts have been invested to include the wind power uncertainty information into power system scheduling problems, such as in stochastic, interval, and robust unit commitment [8]–[17], as well as in probabilistic sizing of operating reserves [18], [19], [20]. Undoubtedly, the performance of the above models is very sensitive to the assumed levels of forecast uncertainty. Generally, if a higher level of uncertainty is considered, the risk in real time operation will be lower, but with additional cost for higher flexibility. Conversely, considering a lower level of uncertainty in advance can decrease the operating cost, but failure to manage uncertainties in real time operation may seriously jeopardize system security. Thus, it is important to select an appropriate uncertainty level of the wind power forecast for power system scheduling and electricity market operation.

Uncertainty intervals have been widely used in interval and robust unit commitment. Interval and robust unit commitment are attractive because its solution is feasible for all outcomes of the uncertainty interval, which is an important aspect from a reliability perspective [10]–[17], although it may result in an overly conservative and costly solution. In the current literature, uncertainty levels are pre-defined before optimizing the power system scheduling problem, and the values of the pre-defined uncertainty levels vary over a wide range. For example, the uncertainty intervals of the wind power forecast are set as $\pm 8\%$ [10], $\pm 10\%$ [11], [14] or $\pm 20\%$ [12] of the wind power forecast value. In [13], $\pm 20\%$ of the installed capacity of the wind farm is considered as the uncertainty range. Reference [17] sets the intervals as the 5- and 95- forecast quantiles. Recently, some research has noticed that the levels of an uncertainty sets have significant effects on system economics and reliability. References [9], [12], [13], [15] and [16] compare the unit commitment cases with different confidence levels or degrees of uncertainty sets. In [21], a fuzzy optimization to minimize costs and risk is proposed. References [22], [23] propose risk-constrained robust unit commitment based on adjustable uncertainty sets of wind power, in which the risk level needs to be pre-determined. Reference [18] presents a hybrid stochastic / interval unit commitment, and optimizes the switch time point between the two formulations. Reference [19] proposes a risk-based optimization method for reserve scheduling. The above research [8]–[20] considers balancing issues between system reliability and economics from different perspectives. However, there has been limited research so far on the optimal selection of wind power forecast uncertainty intervals for power system scheduling and operation.

Furthermore, how to make use of the power system scheduling model with wind power forecast uncertainty information in the electricity market is also a critical problem. Deterministic formulations with dynamic reserves derived from probabilistic wind power forecasts [e.g., 20] have been proposed as an alternative that can be more easily implemented within existing market processes. As to the uncertainty models embedded with wind power forecast uncertainty, limited research has been conducted on the market impacts due to the difficulty in market clearing and price formation under forecast uncertainty. Thus, it is important to investigate the impacts of uncertainty intervals of wind power forecast on the electricity market.

In this paper, we provide a novel model with the aim to optimize the uncertainty intervals of wind power within the system-wide power system scheduling problem. The proposed model is generally derived from the *Interval Unit Commitment with Fixed wind power uncertainty Intervals (F-IUC)* [12], [24], which finds the minimum cost for the central scenario of the wind power (e.g., deterministic wind power forecast) and ensures that the solution is feasible for the full interval representation of uncertainty. Compared to the conventional F-IUC, our model co-optimizes the uncertainty intervals of the wind power forecast and the power system scheduling, obtaining an optimal trade-off between economics and reliability.

The main contributions of this paper are:

- 1) We propose a novel model to optimize the uncertainty intervals of wind power within a power system scheduling model. The proposed model co-optimizes the wind power forecast uncertainty intervals and the day-ahead scheduling problem. We reformulate the model as a MILP problem by linearizing the risk terms (expected energy not served and expect wind curtailment). Hence, the model can directly utilize probability distribution of the wind power forecast without any assumptions on its shape. The proposed method finds the best trade-off between cost and reliability for interval-based unit commitment.
- 2) We apply the proposed model on a two-settlement electricity market based on a modified real-world power system and illustrate significant benefits in terms of reduced costs and more stable prices for energy and reserves.

The rest of the paper is organized as follows. Section II describes the mathematical formulation of the power system scheduling models with fixed and optimal uncertainty intervals of the wind power forecast. The model is then reformulated into a MILP problem. Section III analyzes the model in detail using an illustrative six-bus system. Section IV provides results from a case study in the Illinois power system. Section V concludes this paper.

II. MATHEMATICAL FORMULATION

In this section, the conventional model of F-IUC is firstly introduced. Then we provide the *IUC model with the Optimal Intervals of wind power forecast (OP-IUC)*.

A. F-IUC Model

The F-IUC approach includes operational constraints under three scenarios: 0-the central scenario, which is the forecast or

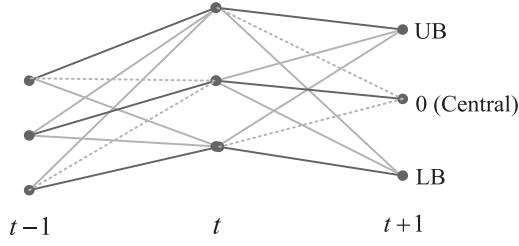


Fig. 1. F-IUC scenarios.

expected scenario of the wind power prediction; UB-the upper bound scenario, when the wind power is at its upper bound of the uncertainty intervals; LB-the lower bound scenario, when the wind power is at its lower bound of the uncertainty intervals. Fig. 1 shows the 0, UB and LB scenarios (in black) and transition scenarios (in grey). Note that the transitions in dotted grey lines are redundant because more serious transitions are represented by the transitions in full grey lines. Thus, the transitions in dotted grey lines are neglected in model formulation.

B. OP-IUC Model

The objective function in (1) has the aim to minimize the operating cost of the central forecast scenario and the costs of the expected energy not served (EENS) and expected wind curtailment (EWC). The objective includes the costs of start-up, shut-down, fuel, energy not served (ENS), wind curtailment (WC), EENS and EWC. ENS and WC are the energy not served and wind curtailment that occurs under the forecast (or central) scenario, which are included to always keep the UC under the central forecast scenario feasible. The ENS and WC under the UB and LB scenarios are not included, because the UC under the UB and LB scenarios are always feasible, due to the fact that the upper and lower bounds are decision variables. EENS and EWC are the expected energy not served and expected wind curtailment induced by wind power forecast uncertainty. The costs of the EWC and EENS can be set based on different risk preferences. Note that the cost of wind curtailment (WC and EWC) can be interpreted as part of a subsidy for wind power, e.g., such as the production tax credit for wind in the US. The production tax credit is a subsidy scheme where wind power producers receive a tax credit for every MWh of generation. In effect, they are therefore willing to produce even if the energy price is negative. From a modeling perspective, this is equivalent to a negative production cost for wind power or a cost of wind power curtailment. In this paper, the probability density functions (PDFs) of the wind power forecast error are assumed to be known, and based on a probabilistic forecast.

$$\begin{aligned}
 \min \quad & \sum_{t=1}^{NT} \sum_{i=1}^{NG} \pi^c \cdot F_{ci} \cdot I_{it} + SU_{it} + SD_{it} \\
 & + \sum_{t=1}^{NT} \sum_{b=1}^{NB} \pi^{ens} P_{D,bt}^{ens} + \sum_{t=1}^{NT} \sum_{j=1}^{NW} \pi^{wc} P_{W,jt}^{wc} \\
 & + \sum_{t=1}^{NT} \sum_{j=1}^{NW} (\pi^{ens} P_{jt}^{ens} + \pi^{wc} P_{jt}^{wc}) \quad (1)
 \end{aligned}$$

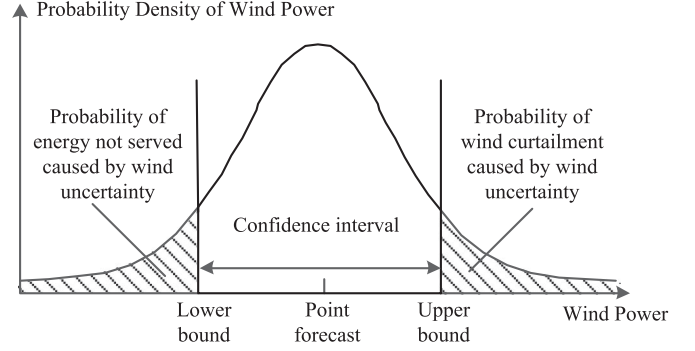


Fig. 2. EENS and EWC illustration.

where: $F_{ci} = aP_{G,it}^0 + bP_{G,it}^0 + c$, $SU_{it} = \pi^{st} \alpha_{it}$, $SD_{it} = \pi^{sd} \beta_{it}$.

Fig. 2 illustrates the EENS and EWC concepts in the OP-IUC formulation. According to the IUC concept, when wind power generates above the upper bound of the wind power forecast interval, the extra wind power needs to be curtailed. When the wind power generates below the lower bound of the wind power forecast interval, the resulting unmet load needs to be shed. The EENS and EWC are modeled using the integration of the probability distribution of the wind power forecast, as shown in (2) and (3), where f_p is the probability distribution function of the wind power forecast, $\omega_{jt}^{LB} P_{W,jt}^{inst} = P_{W,jt}^{LB}$,

$$P_{jt}^{ens} = \int_{x=0}^{x=\omega_{jt}^{LB}} (\omega_{jt}^{LB} - x) \cdot f_p(x, P_{W,jt}^0) dx \quad (2)$$

$$P_{jt}^{wc} = \int_{x=\omega_{jt}^{UB}}^{x=1} (x - \omega_{jt}^{UB}) \cdot f_p(x, P_{W,jt}^0) dx \quad (3)$$

OP-IUC is subject to the constraints (4)–(11) discussed below. Constraint (4) determines if generator i is started up or shut down at time t . Constraints (5) and (6) enforce minimum up and down time for the remaining time intervals. Equation (7) enforces the nodal power balance. Constraint (8) is the DC power flow constraint. Constraint (9) enforces the minimum and maximum generator output limits. Constraint (10) enforces the up and down ramp limits for 0, UB and LB scenarios and transition scenarios. Here for simplification, we assume all re-dispatch is undertaken by all committed units, except for wind power

$$\alpha_{it} - \beta_{it} = I_{it} - I_{i(t-1)} \quad \forall t, i \quad (4)$$

$$\begin{aligned}
 I_{it} - I_{i(t-1)} &\leq I_{ik} \quad \forall t \geq 2, \forall k \\
 &= t, \dots, \min(NT, t + T_i^{on} - 1) \quad (5)
 \end{aligned}$$

$$\begin{aligned}
 I_{ik} &\leq 1 + I_{it} - I_{i(t-1)} \quad \forall t \geq 2, \forall k \\
 &= t, \dots, \min(NT, t + T_i^{off} - 1) \quad (6)
 \end{aligned}$$

$$\sum_{i=1}^{NG} P_{G,it}^s + \sum_{j=1}^{NW} P_{W,jt}^s = \sum_{b=1}^{NB} (P_{D,bt}^f - P_{D,bt}^{ens}) \quad \forall t \forall s \quad (7)$$

where $P_{W,jt}^0 = P_{W,jt}^f - P_{W,jt}^{wc}$

$$-PL_{l,\max} \leq \sum_{b=1}^{NB} G_{l-b} P_{D,bt} + \sum_{i=1}^{NG} G_{l-i} P_{G,it}^s + \sum_{j=1}^{NW} G_{l-j} P_{W,jt}^s \leq PL_{l,\max} \quad \forall t, l \quad (8)$$

$$P_{Gi,\min} I_{it} \leq P_{G,it}^s \leq P_{Gi,\max} I_{it} \quad \forall t, i \quad (9)$$

$$-I_{it} R D_i - \beta_{it} R S D_i \leq P_{G,it}^{s'} - P_{G,i(t-1)}^{s''} \leq I_{i(t-1)} R U_i + \alpha_{it} R S U_i \quad (10)$$

$$\forall (s', s'') \in \left\{ (UB, UB), (LB, LB), (0, 0), (UB, LB), (LB, UB), (UB, 0), (LB, 0) \right\}.$$

It should be noted that the transmission constraint (8) may not cover the worst possible scenario if there are multiple wind farms at different locations in the transmission network. Although such worst scenarios seldom happen, we can add (11) and (12) if necessary, which represent the tightest constraints. For a similar approach we refer to [12]

$$-PL_{l,\max} \leq \sum_{b=1}^{NB} G_{l-b} P_{D,bt} + \sum_{i=1}^{NG} (G_{l-i}^+ P_{G,it}^{UB} + G_{l-i}^- P_{G,it}^{LB}) + \sum_{j=1}^{NW} (G_{l-j}^+ P_{W,jt}^{UB} + G_{l-j}^- P_{W,jt}^{LB}) \leq PL_{l,\max} \quad \forall t, l \quad (11)$$

$$-PL_{l,\max} \leq \sum_{b=1}^{NB} G_{l-b} P_{D,bt} + \sum_{i=1}^{NG} (G_{l-i}^- P_{G,it}^{UB} + G_{l-i}^+ P_{G,it}^{LB}) + \sum_{j=1}^{NW} (G_{l-j}^- P_{W,jt}^{UB} + G_{l-j}^+ P_{W,jt}^{LB}) \leq PL_{l,\max} \quad \forall t, l \quad (12)$$

where the “+” and “-” mean the positive and negative parts of the matrix, respectively.

Compared to the deterministic UC, the OP-IUC considers the transmission constraints under the baseline forecast scenario as well as under the LB and UB scenarios. Thus, OP-IUC maintains a higher capacity margin in the transmission lines, which helps to prevent line overflows and cascading failure problems.

C. Linearization of OP-IUC

The key problem of the proposed model is that the wind power intervals, i.e., ω_{jt}^{LB} and ω_{jt}^{UB} are decision variables, and are given in the form of integration terms as part of the objective function. Their values are not known prior before solving the unit commitment.

For the risk terms P_{jt}^{pens} and P_{jt}^{wc} , it can be proven that their second order partial derivative matrices are non-negative, which not depends on the shape of the probability function. Thus, they are convex and can be piecewise linearized as in [18] and [25]. For each risk term, the parameters of each segment need to be calculated separately. In this paper, we provide a straightforward method to linearize P_{jt}^{pens} and P_{jt}^{wc} . Firstly, the ranges of the

integration are extended to $[0, 1]$ to remove the decision variables ω_{jt}^{LB} and ω_{jt}^{UB} from the integral limits in (2) and (3). Note that ω_{jt}^{LB} and ω_{jt}^{UB} come out as the endpoints of the integral limits, which indicate that the integrand functions $\omega_{jt}^{LB} - x$ and $x - \omega_{jt}^{UB}$ only work when they are non-negative. Thus (2) and (3) can be transformed into (13) and (14), respectively,

$$P_{jt}^{pens} = \int_{x=0}^{x=1} [\omega_{jt}^{LB} - x]^+ \cdot f_p(x, P_{W,jt}^0) dx \quad (13)$$

$$P_{jt}^{wc} = \int_{x=0}^{x=1} [x - \omega_{jt}^{UB}]^+ \cdot f_p(x, P_{W,jt}^0) dx \quad (14)$$

where $[t]^+ = \max\{t, 0\}$

The next step is to transform (13) and (14) into discrete forms, which follows the linearization method of conditional value-at-risk (CVaR) in [26] and [27]. Dummy continuous variables ξ_{mjt} and ζ_{mjt} are added to replace $[\omega_{jt}^{LB} - x]^+$ and $[x - \omega_{jt}^{UB}]^+$, respectively. The probability distribution of the wind power forecast $f_p(x, P_{W,jt}^0)$ is discretized to f_{mjt} . If the original data of the probability distribution is discrete, it can be utilized directly. Due to the factors that probability is intrinsically non-negative and the objective function is to minimize P_{jt}^{pens} and P_{jt}^{wc} , the (13) and (14) can be transformed into (15)–(17) and (18)–(20), respectively. Such a transformation is not based on any assumption on the shape of the probability distribution. Unsymmetrical, non-smooth, non-parametric, or discrete distributions are all applicable

$$P_{jt}^{pens} \geq \sum_{m=1}^M \xi_{mjt} f_{mjt} \quad (15)$$

$$\xi_{mjt} \geq 0 \quad \forall m, j, t \quad (16)$$

$$\xi_{mjt} \geq \omega_{jt}^{LB} - x_m \quad \forall m, j, t \quad (17)$$

$$P_{jt}^{wc} \geq \sum_{m=1}^M \zeta_{mjt} f_{mjt} \quad (18)$$

$$\zeta_{mjt} \geq 0 \quad \forall m, j, t \quad (19)$$

$$\zeta_{mjt} \geq x_m - \omega_{jt}^{UB} \quad \forall m, j, t. \quad (20)$$

Finally, the complete OP-IUC model is then formulated as: (1), (4)–(12), and (15)–(20). Compared to the previously mentioned linearization method in [18], the proposed method uses the discrete probability distribution of the wind power forecast directly and avoids complex calculations for linearization.

D. Market Application Model (m.)

We apply uncertainty intervals as reserve requirements for wind power in a two settlement electricity market, showing that how to apply the OP-IUC results and how it performs in day-ahead (DA) and real-time (RT) markets. The overall structure of the simulated market operation resembles U.S. electricity markets with a DA market and a RT market, which are cleared in sequence.

Generally, both the DA and RT markets are cleared by running a market clearing model, as shown in Fig. 3. When conducting

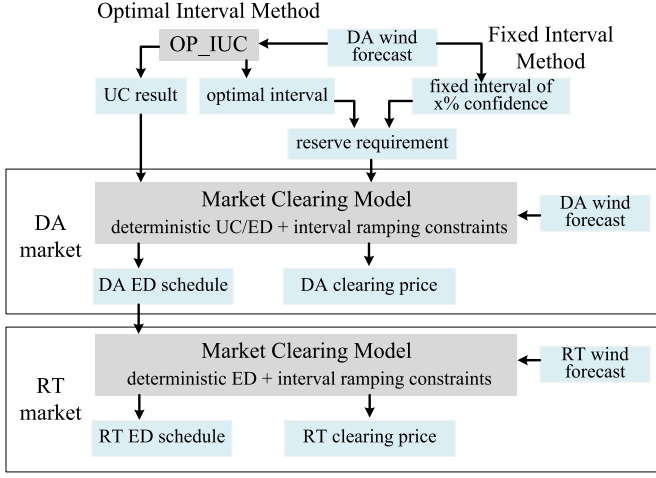


Fig. 3. Market application process.

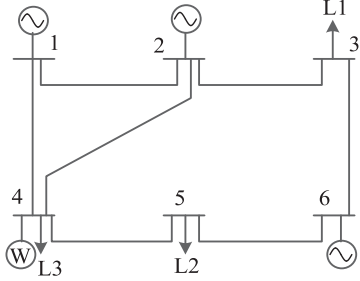


Fig. 4. Six-bus system.

the market application analysis, first run the OP-IUC model or select a fixed confidence interval of wind power forecast to obtain the inputs of the DA market clearing model. Then run the DA and RT market clearing models in sequence to obtain the generation schedules and prices. The prices for energy and reserves are derived from the energy and reserve balance constraints.

Note that the inputs and formulation of the DA market clearing model are different in the OP-IUC and Fixed Interval methods, which are explained in the followings.

OP-IUC method: the DA market clearing model is a deterministic economic dispatch (ED) model including interval ramping constraints, in which the UC are the results from the OP-IUC model, and the reserve requirement is determined by the optimal wind power interval.

Fixed Interval method: the DA market clearing model is a deterministic UC model including interval ramping constraints, in which the reserve requirement is determined by the selected confidence interval of the wind power forecast. When calculating the market clearing price, the UC results are fixed in order to derive the shadow prices from the resulting ED model.

For both methods, the RT market clearing model is a deterministic ED model. In this paper, the flexibility of the system operator in terms of addressing DA forecasting errors (e.g., commitment of additional units between DA and RT) is not considered.

The full market clearing model formulation is presented below.

Objective function: minimize the cost of generation, ENS, and WC

$$\min \sum_{t=1}^{NT} \sum_{i=1}^{NG} \pi^c \cdot F_{ci}(P_{G,i,t}^0) \cdot I_{it} + SU_{it} + SD_{it} + \sum_{t=1}^{NT} \sum_{b=1}^{NB} \pi^{ens} P_{D,b,t}^{ens} + \sum_{t=1}^{NT} \sum_{j=1}^{NW} \pi^{wc} P_{W,j,t}^{wc}. \quad (21)$$

Constraints: (4)–(10), only the forecast (central) scenario is included; (22) and (23) are system reserve requirement constraints; (24) and (25) are limit reserve capacity for individual units; (26) and (27) interval ramping constraints for individual units. Note that in the case study of this paper, we assume the wind power forecast in the RT market is perfect, and the reserve requirement in the RT market is set as zero. Under such assumptions, (22)–(27) can be neglected in the RT market clearing model.

$$\sum_{i=1}^{NG} R_{G,i,t}^{up} \geq \hat{R}_{req,t}^{up} \quad \forall t \quad (22)$$

$$\sum_{i=1}^{NG} R_{G,i,t}^{dn} \geq \hat{R}_{req,t}^{dn} \quad \forall t \quad (23)$$

$$0 \leq R_{G,i,t}^{up} \leq P_{G,i,max} I_{it} - P_{G,i,t}^0 \quad \forall t, i \quad (24)$$

$$0 \leq R_{G,i,t}^{dn} \leq P_{G,i,t}^0 - P_{G,i,min} I_{it} \quad \forall t, i \quad (25)$$

$$P_{G,i,t}^0 + R_{G,i,t}^{up} - P_{G,i,t-1}^0 + R_{G,i,t-1}^{dn} \leq I_{i(t-1)} RU_i + \alpha_{it} RSU_i \quad \forall t, i \quad (26)$$

$$P_{G,i,t-1}^0 + R_{G,i,t-1}^{up} - P_{G,i,t}^0 + R_{G,i,t}^{dn} \leq I_{it} RD_i + \beta_{it} RSD_i \quad \forall t, i. \quad (27)$$

In the above model, the parameters of reserve requirements ($\hat{R}_{req,t}^{up}$ and $\hat{R}_{req,t}^{dn}$) are determined by the OP-IUC optimization results, which are calculated by

$$\hat{R}_{req,t}^{up} = \sum_{j=1}^{NW} (P_{W,j,t}^0 - P_{W,j}^{inst} \omega_{jt}^{LB}) \quad \forall t \quad (28)$$

$$\hat{R}_{req,t}^{dn} = \sum_{j=1}^{NW} (P_{W,j}^{inst} \omega_{jt}^{UB} - P_{W,j,t}^0) \quad \forall t. \quad (29)$$

For the above model, the local marginal price (LMP) can be calculated by adding up the energy price (i.e., market clearing price, MCP) and the congestion component. Energy price is derived as the shadow price of the energy balance constraint (7), and the congestion component can be calculated by multiplying the shift factor and the shadow price of the transmission network constraints (8) [28]. The prices of the up and down reserves are derived as the shadow prices of the up and down reserve requirement constraints (22) and (23). Note that in practical power system operation, operators maintain other resources to avoid systematic risks, such as regulation reserve and non-spinning reserves, and demand response. In this model, only spinning reserves are included for simplification.

TABLE I
GENERATOR DATA

Unit	Pmax (MW)	Pmin (MW)	Ini. State (h)	Min Off(h)	Min On (h)	Up/Dn Ramp (MW)	Startup/Shut down Ramp (MW)
G1	220	90	4	-4	4	40	90
G2	100	20	2	-3	2	20	20
G3	30	10	-1	-1	1	15	15

Unit	Fuel consumption Function			Start up Fuel (MBtu)	Shut down Fuel (MBtu)	Fuel Price (\$)
	a	b	C (MBtu/ MW ² h)			
G1	176.9	13.5	0.0004	180	50	1.2469
G2	129.9	32.6	0.001	360	40	1.2461
G3	137.4	17.6	0.005	60	0	1.2462

TABLE II
TRANSMISSION LINE DATA

Line No.	From Bus	To Bus	X (p.u)	Flow Limit (MW)	Line No.	From Bus	To Bus	X (p.u)	Flow Limit (MW)
1	1	2	0.17	200	5	2	3	0.037	200
2	1	4	0.258	100	6	4	5	0.037	200
3	2	4	0.197	100	7	3	6	0.018	200
4	5	6	0.14	100					

In order to distinguish the simulation results of the market clearing models, we name them ended by “m.”. Thus, in the following context, the market clearing results of the OP-IUC and Fixed Interval methods are marked by “OP-IUC m.” and “x% m.”, respectively.

III. AN ILLUSTRATIVE EXAMPLE

In this section, we present numerical experiments carried out on a modified six-bus system [8] with one wind farm to show the effectiveness of the proposed model and algorithm. The generating units’ cost curves are 4-piecewise linear functions, and the costs of EENS and EWC are represented as linear functions with 100 piecewise segments. The experiments are performed on a PC with Intel(R) Core(TM) i5 2.60 GHz and 8 GB of memory. The algorithms are implemented in MATLAB and programmed using YALMIP [29]. The MILP solver is CPLEX 12.6 [30]. The optimality gap is set as $1e-6$. The solution time for the one day simulation of OP-IUC is 5.57 s.

A. Six-Bus System

The six-bus system depicted in Fig. 4 is used to illustrate the proposed method. The system has three thermal units located in No.1, No.2 and No.6 buses, one wind farm located in No.4 bus, and 7 transmission lines [8]. The characteristic of the thermal units and transmission lines are listed in Table I and Table II, respectively. The installed capacity of the wind farm is 100 MW. The load curves, day-ahead and real-time forecasts of wind generation, and the wind power probability distribution are all

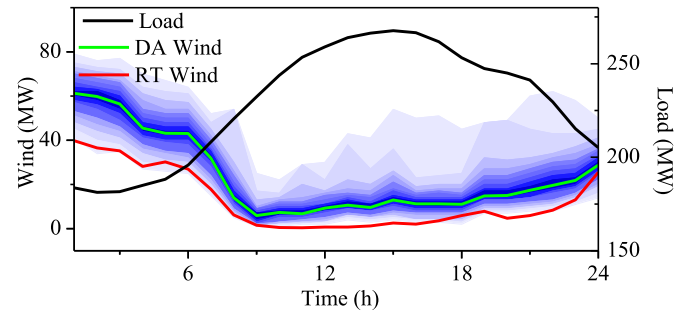


Fig. 5. Data of load and wind power (July 28th).

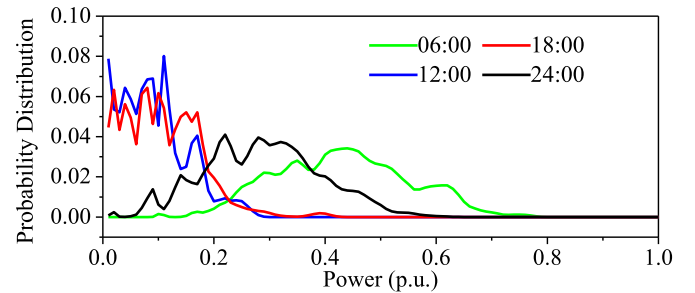


Fig. 6. Probability distributions of wind power forecast (July 28th).

scaled down from the data of the U.S. Illinois power system. Detailed data information refers Section IV part A. One day (July 28th) simulation is presented in this section to show the effectiveness of the proposed method. The daily load, day-ahead and real-time wind power curves are shown in Fig. 5. The mean absolute error (MAE) of the day-ahead wind power forecast on July 28th is 10.95% of the installed capacity. The confidence intervals of wind power forecast uncertainty from 10% to 90% confidence levels are highlighted in blue. The probability distributions of the wind power forecast for four selected time periods are shown in Fig. 6. The figure illustrates that the shapes of the distributions vary extensively between the four hours.

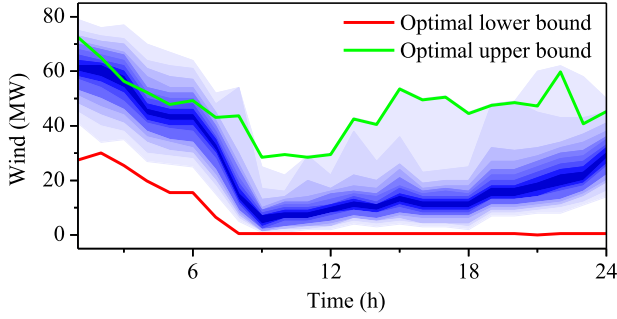


Fig. 7. Optimal uncertainty intervals of wind power (July 28th).

TABLE III
COMPARISON WITH PREDEFINED INTERVAL CASES (JULY 28TH)

Confidence level	Cost (\$)			
	Gen	EENS	EWC	Total ¹
10%	100342	262730	3419	366491
20%	100415	214961	2801	318177
30%	100471	180263	2247	282982
40%	100555	140969	1702	243226
50%	101395	111546	1277	214218
60%	101488	82036	914	184438
70%	102133	56765	541	159439
80%	104264	36951	231	142401
90%	108126	21002	61	129189
optimal	103289	268	857	104414

¹The total cost includes the costs of generation, EENS, EWC, ENS and WC. The cost of ENS under 80% confidence is \$954.92. The costs of ENS and WC under other confidence levels are all zero, thus they are not shown in the table.

The prices of the load shedding (ENS and EENS) and wind curtailment (WC and EWC) are set as 3500\$/MWh [20] and 50\$/MWh [23], respectively.

B. Cost Effectiveness

The optimized objective is \$104414, including the cost of generation \$103289, the cost of EENS \$268, EWC \$857, ENS \$0, and WC \$0. The optimized upper and lower bounds of the wind power uncertainty interval are shown in Fig. 7. The figure shows that the optimal intervals are generally shifted downward compared to the symmetric intervals because the costs of ENS/EENS are much higher than that of WC/EWC. The confidence levels of the optimal interval vary at different time periods. For instance, the optimal upper bound is within the 10% confidence intervals in hour 3, while it is above the 90% interval in hour 10. Furthermore, note that since the objective of the OP-IUC model is to minimize the total cost of operation and risk, the optimal intervals are the widest possible and reflect the system's ability of accommodating wind power under a certain optimal solution for UC and ED.

C. Comparison to F-IUC Under Predefined Interval

The results are compared with F-IUC under predefined uncertainty intervals of wind power, ranging from 10% to 90% confidence intervals in Table III. With the increase in the

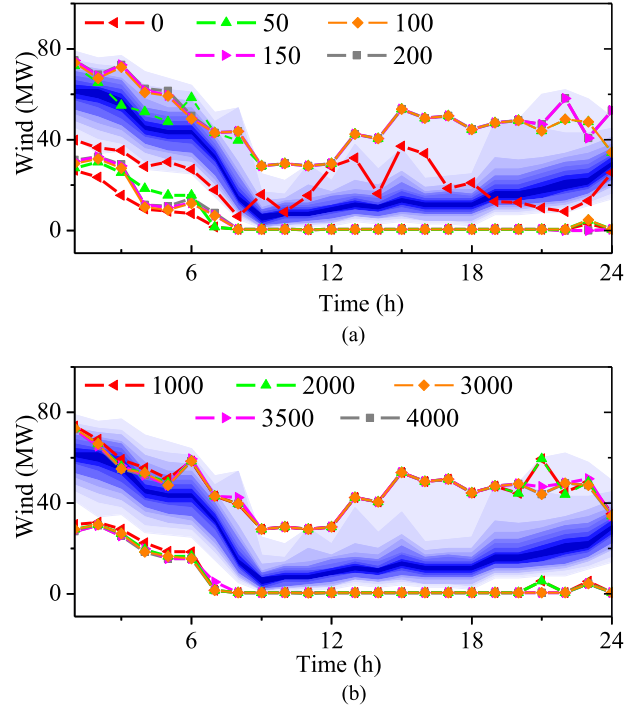


Fig. 8. Optimal uncertainty intervals under different costs of (a) WC/EWC (b) ENS/EENS (July 28th).

confidence levels, the operating cost increases while the costs of EENS and EWC decrease. In this case, the total cost seems to decrease monotonically with wider intervals. However, this does not mean the best solution is obtained under the widest intervals. Actually, the F-IUC models under the confidence intervals of 95% and 99% are infeasible.

Compared with the conventional predefined interval UC, the OP-IUC schedule accommodates relatively wide wind power uncertainty intervals, while at the same time maintaining a low generation cost. The thermal generation cost only increases by 2.94% compared with the lowest cost (10% case), while the cost of the risk decreases by 94.66% compared with the lowest risks (90% case). Thus the OP-IUC reaches the optimal trade-off between operation costs and risk costs.

D. Sensitivity Analysis of WC/EWC and ENS/EENS Costs

Optimal uncertainty intervals under different costs of WC/EWC and ENS/EENS are shown in Fig. 8. Fig. 8(a) shows that the lower bounds are not significantly sensitive to the WC/EWC price; except that the upper bound in the \$0/MWh case is much lower than that of the other cases. Fig. 8(b) shows that both the lower and upper bounds are not significantly sensitive to the ENS/EENS costs. The main difference is that the lower bound in the \$1000/MWh and \$2000/MWh case is a little higher than that of the other cases.

E. Market Application Analysis

Traditionally, spinning reserves for wind power are scheduled using deterministic and static criteria, e.g., by a fixed amount or a fixed ratio of the wind power forecast, or from quantile

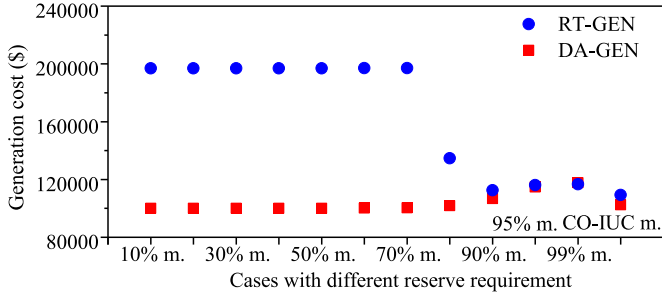


Fig. 9. DA and RT Generation cost in market cases (July 28th).

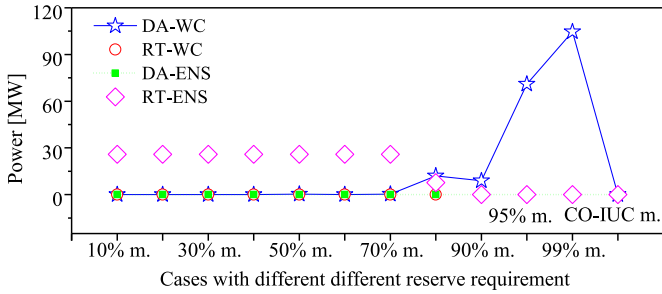


Fig. 10. DA and RT WC and ENS in market cases (July 28th).

intervals based on the probability distribution of wind power forecast errors. In this paper, the proposed optimal uncertainty intervals of wind power provide an alternative approach to estimate reserve requirements for wind power that better reflect the trade-off between system cost and reliability.

The OP-IUC m. case results are compared to those of the deterministic UC at the DA stage with 10% to 90%, 95% and 99% quantile reserve requirements. The cases with x% quantile reserves are referred as x% m. case. Note that all these cases have the same amount reserve requirements in the RT market. Fig. 9 shows that the generation cost in the OP-IUC m. cases are relatively lower compared to other cases in both the DA and RT market. Moreover, the cost is higher when the reserve requirement is higher in the DA market, because the system needs more flexibility. In the RT market, the costs in the 10% to 80% m. cases are higher than in the 90% m. case. This is because the lack of reserves results to high penalty costs in the RT market. The costs in the 95% and 99% m. cases are also higher than for case 90% m, indicating that more reserves are maintained than necessary, leading to inefficient dispatch.

The WC and ENS are given in Fig. 10. It can be seen that the WC and ENS are zero in the OP-IUC m. case. As for the other results, the DA WC is larger in 80% to 99% m. cases, because the reserve requirement is large, which squeezes the room for wind power. Under such situation, integrating wind power will largely increase the total cost. The RT ENS is larger in the 10% to 80% cases because the RT wind is less than forecast, and the lack of reserve leads to load shedding in RT.

Three typical cases, i.e., 20% m., 90% m. and OP-IUC m., are selected to study the impacts of different scheduling strategies on prices. Table IV shows the average energy prices and their standard deviation (STD) along with average reserve prices.

TABLE IV
AVERAGE ENERGY AND RESERVE PRICES (28TH JULY)

	OP-IUC m.		20% m.		90% m.	
	DA	RT	DA	RT	DA	RT
Energy Price (\$)	24.5	26.5	25.3	435.5	24.5	26.0
STD Energy Price (\$)	11.9	11.9	11.4	932.4	41.5	12.1
RSV UP Price (\$)	0.0	0.0	0.0	0.0	10.7	0.0
RSV DN Price (\$)	0.4	0.0	0.0	0.0	12.4	0.0

TABLE V
GENERATION SCHEDULES HOURS 6 AND 7

Power/Hour	t = 6			t = 7		
	G1	G2	G3	G1	G2	G3
$P_{G,it}^{LB}$ (MW)	152.1	0.0	25.0	168.1	0.0	25.0
$P_{G,it}^0$ (MW)	149.0	0.0	10.0	166.1	0.0	10.0
$P_{G,it}^{UB}$ (MW)	128.1	0.0	10.0	146.1	0.0	10.0
$P_{W,it}^{wc}$ (MW)		6.3			0.0	
$P_{W,it}^{ens}$ (MW)		0.0			0.0	
$R_{req,t}^{up}$ (MW)	18.1			17.0		
$R_{req,t}^{dn}$ (MW)	20.9			20.0		
Load (MW)	195.8			208.1		
Wind (MW)	43.1			32.0		

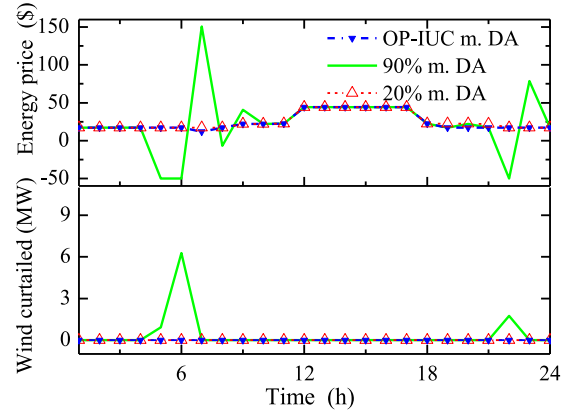


Fig. 11. Average energy prices and wind curtailed in market cases (July 28th).

Note that the energy prices analyzed here do not include the congestion components. The STDs indicate that the volatility of the energy prices is much lower in OP-IUC m. cases in both the DA and RT market. Especially, the 20% m. case has higher RT energy prices due to the penalty cost of ENS, and the 90% m. case has higher DA reserve prices.

Fig. 11 shows the energy prices and wind curtailment in the DA market. It is interesting to note that wind curtailment is often accompanied with large energy price variations. The reason is that the interval ramping constraints couples the prices of adjacent hours. For instance, Table V gives the generation schedules in hour 6 and hour 7. If the load increases by 1 MW in hour 7, one of the thermal generators (G1, G2, or G3) has to increase its generation by 1MW. In this case the marginal unit is G1, and $P_{G1,t=7}^0$ needs to increase from 166.1 MW to

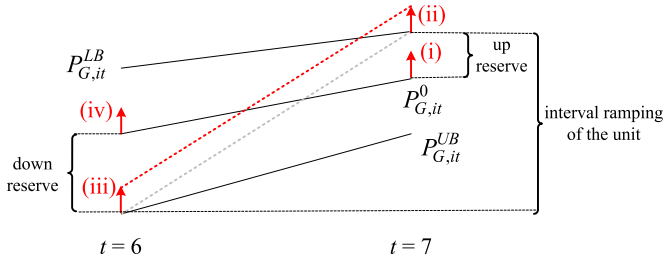


Fig. 12. Impact of interval ramping constraint of G1 on its dispatch in hours 6 and 7 for base (0), upper (UB), and lower (LB) wind forecast scenarios.

167.1 MW, as shown by arrow (i) in Fig. 12. Meanwhile, the up reserve requirement needs to be maintained. The G1 dispatch in scenario LB ($P_{G,7}^{LB}$) has to increase from 168.1 MW to 169.1 MW, as shown by arrow (ii). Considering interval ramping constraints of the unit, the G1 generation in scenario UB in hour 6 ($P_{G,6}^{UB}$) is coupled with $P_{G,7}^{LB}$, and the unit ramping capability is 40 MW/h, i.e., $-40 \leq P_{G,7}^{LB} - P_{G,6}^{UB} \leq 40$. Thus, the $P_{G,6}^{UB}$ has to increase from 128.1 MW to 129.1 MW, as shown by arrow (iii). Moreover, in order to keep the down reserve requirement of the power system, the thermal generators (G1, G2 or G3) have to increase 1 MW in scenario 0. In this case $P_{G,6}^0$ increases from 149.0 MW to 150.0 MW, as shown by arrow (iv). Note that there is wind power curtailed in hour 6. The 1MW generation increase of G1 in period 6 results in 1 MW more wind power curtailment. Due to the wind power curtailment penalty mechanism, the objective value increases due to (a) the change of generation cost in hour 7 and (b) the increase of wind curtailment penalty cost in hour 6. Thus, the energy price, which is the system marginal cost, in hour 7 spikes to 150.9\$/MWh (Fig. 11).

IV. ANALYSIS ON A REAL POWER SYSTEM

In this section, we apply the optimal uncertain interval UC method to the Illinois power system in the U.S. The experiments are performed on Linux-based server at Argonne National Laboratory with 2x Intel E5430 Xeon CPUs and 32 GB of RAM. The algorithms are implemented on AMPL/CPLEX. The optimality gap was set at $5e-4$. The average solving time is 1191s for a one-day OP-IUC simulation.

A. Simplified Illinois Power Grid

A simplified representation of the power system in the U.S. state of Illinois is used to study the potential effects of unit commitment and economic dispatch using optimal intervals of wind power uncertainty under a large-scale expansion of wind power. The data set is based on [31]. The system consists of 210 thermal plants, with total capacity of 41380 MW. The analysis period is July–October 2006, i.e., 123 days. The installed capacity of wind power is assumed as 14000 MW in a future scenario when 20% of the load (including export) is met by wind. The load profile is based on the historical 2006 data in the state of Illinois. The DA and RT wind power forecast are based on the wind speed forecast by aggregating data for 15 hypothetical sites in Illinois in 2006 [20]. The MAE of the wind power forecast is

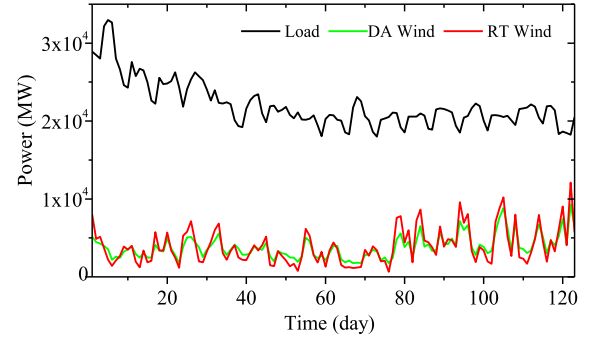


Fig. 13. Daily average data of the Illinois power system (123 days).

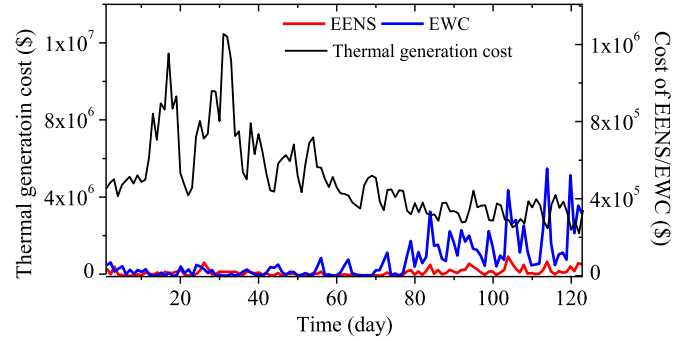


Fig. 14. Cost of generation, cost of EENS and EWC (case OP-IUC 123 days).

7.6% of the install capacity. The daily average data of load and wind power are plotted in Fig. 13. The probability distribution of the wind power forecast is a 100-segment discrete distribution, which is based on a quantile-copula kernel density estimator (KDE) using a set of historical data. The historical predicted and realized wind power data is the input of the KDE method. For more details about the advantages and implementation of the probabilistic wind power forecasting approach, we refer to [32] and [33].

B. OP-IUC System Cost Results

The costs of generation, EENS and EWC in 123 days from the OP-IUC model are shown in Fig. 14. The costs of EWC are high in the later months, because the load is relatively lower and the wind power is higher in September and October, making it more difficult to accommodate all the wind power.

The risk costs of EENS and EWC under optimal and predefined 90%, 95% and 99% confidence intervals are compared in Fig. 15. It can be observed that the flexible uncertainty intervals in OP-IUC contribute to a significant lower risk cost, which is even much lower than that under 99% confidence intervals. Fig. 16 shows the ratio of the cost of EENS and EWC vs. cost of generation in OP-IUC. In general, the risk cost incurred by wind power is relatively low, or only a small portion of the operation cost. Based on this assumption, some risk-based research limits the risk cost to a predefined threshold in the deterministic or chance constraints for UC and ED problems [22], [23]. However, the results in this study show that the risk cost of EENS and EWC can be more than 20% of the generation cost in some

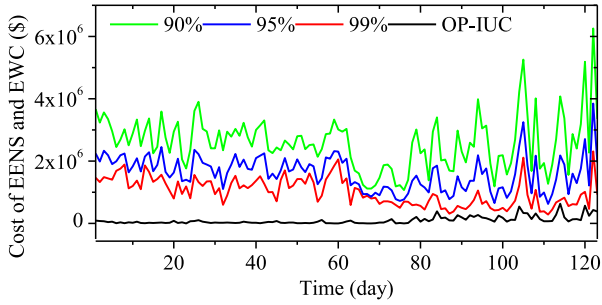


Fig. 15. Risk cost under different intervals (123 days).

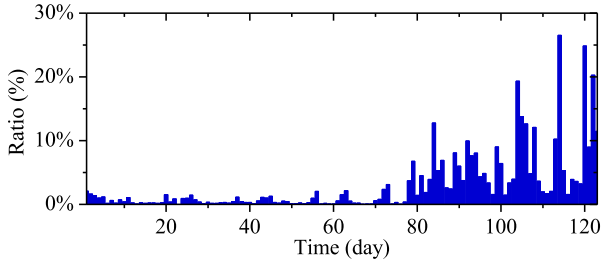


Fig. 16. EENS and EWC cost vs. generation cost (case OP-IUC 123 days).

TABLE VI
AVERAGE DAILY ED COSTS (123 DAYS)

		Cost (\$)			
		Gen	ENS	WC	Total
DA	OP-IUC m.	4919370	0	716	4920086
	20% m.	4859089	0	9490	4868578
	90% m.	5096938	0	181647	5278585
RT	OP-IUC m.	5000178	8147	16838	5025163
	20% m.	4923092	8408310	9130	13339112
	90% m.	5078850	246670	20282	5345802

cases, and the ratio varies largely day by day. From this point, the OP-IUC method can find the optimal tradeoff of the system operation and risk costs, as well as provide an improved benchmark to determine the risk threshold (or ratio) for other risk-based UC models.

C. Market Application Analysis

Similar to Section III-E, we also apply the optimal uncertainty intervals as reserve requirements for wind power on the Illinois case. The daily average ED costs in three typical cases, i.e., 20% m., 90% m. and OP-IUC m., are shown in Table VI. Table VII shows the average reserve prices, energy prices and standard deviations (STD). They show that in the DA market, the total cost in OP-IUC m. case is 6.8% less than in the 90% m. case, and 1.1% larger than in the 20% m. case. In the RT market, the total cost in OP-IUC m. case is 6.0% less than in the 90% m. case, and 62.3% less than in the 20% m. case. Meanwhile, note that the 20% m. case has a very high amount of ENS and therefore a high energy price in the RT market, because the 20%

TABLE VII
AVERAGE ENERGY AND RESERVE PRICES (123 DAYS)

	OP-IUC m.		20% m.		90% m.	
	DA	RT	DA	RT	DA	RT
Energy Price(\$)	15.5	27.7	19.0	614.5	14.1	93.1
STD Energy Price(\$)	15.8	203.4	49.1	1312.8	77.2	511.7
RSV UP Price(\$)	0.6	0.0	2.2	0.0	10.8	0.0
RSV DN Price(\$)	0.8	0.0	1.4	0.0	14.2	0.0

m. case schedule insufficient resources for RT operation. The OP-IUC m. case has the least ENS since the OP-IUC generally schedules a lower uncertainty interval to cover those situations. As to the generation cost, the costs in OP-IUC m. case are 3.5% and 1.5% less than the 90% m. case in the DA and RT markets, and 1.2% and 1.6% larger than the 20% m. case in DA and RT market, respectively. Meanwhile, the OP-IUC m. case has the lowest energy price STDs in both DA and RT market, indicating less volatile energy prices. Overall, it can be seen that the amount of reserve requirement substantially influences the price, since the same resources usually provide both energy and reserves. The OP-IUC provides an alternative to determine an appropriate amount of reserves in market operation, which reduces the operation risks and price variations.

From the perspective of wind power producers, different forecast intervals in the market clearing influence their revenues. Under a narrow interval, the reserve requirement of the power system may be insufficient. When the wind power is over-forecast or under-forecast, the power system may need to shed the load (over-forecast) or curtail wind power (under-forecast), which leads to very high (over-forecast) or negative (under-forecast) energy prices in the RT market. Under such situations, wind the power producers have to buy back energy at scarcity prices or sell wind power at negative prices in the RT market, which causes revenue losses. Similarly, under wide forecast intervals, the reserve requirement may be excessive, which may lead to wind curtailment in the DA market, and further result in negative energy price in the DA market due to the assumed curtailment penalty. Under such situations, wind power producers have to sell energy at negative prices in the DA market, which also causes revenue losses. Overall, very narrow or very wide intervals tend to bring high volatility in energy prices and therefore result in high risk of revenue loss. From this point of view, the optimal interval stabilizes the energy prices, as shown in Table VII, and therefore also stabilizes revenue and reduces risk of losses for the wind power producers.

V. CONCLUSION

This paper proposes a method to optimize the uncertainty intervals of wind power in interval unit commitment. Compared to the conventional interval scheduling model with predefined fixed intervals, the proposed model achieves a better tradeoff between economics and reliability.

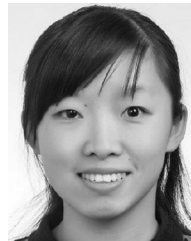
The proposed model is tested on a two-settlement electricity market based on a six-bus test systems and a simplified

representation of the power system for the state of Illinois. The results show that the proposed method can help ISOs to determine an appropriate amount of reserve requirements, which can benefit the electricity market by decreasing the risks of wind power curtailment and load shedding, and further helping to produce more stable prices with less volatility in both the DA and RT markets. Meanwhile, the model is computationally efficient, which ensures the applicability in large-scale power systems with uncertainties.

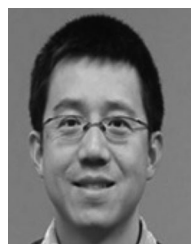
In future work, we plan to investigate optimal uncertainty representations under other scheduling strategies, like robust or stochastic unit commitment. Moreover, we may further consider other risks (e.g., cascading failures) and extend our linearization method to other risk quantifications (e.g., risks of transmission line overflows). On the market side, we will investigate a broader set of market implications from different operational strategies. Other interesting directions for future research include the potential consideration of the operators' risk preferences in risk-based operational strategies.

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